QP Code: D 112416	Total Pages: 2	Name:
		Register No.

FIRST SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2024

(CUFYUGP)

APH1MN102/PHY1MN102: Properties of matter and Thermodynamics

2024 Admission onwards

Maximum Time :2 Hours | **Maximum Marks :70**

Section A

All Questions can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Check the condition of static equilibrium in the case of a thrown baseball that does not rotate as it sails through the air
2	Define the Center of gravity and center of mass. How are they related
3	Distinguish elastic and plastic material with reference to elastic limit
4	If a metal wire has its length doubled and its diameter tripled, by what factor does Young's modulus change?
5	State and explain Pascal's law?
6	An ice cube floats in a glass of water. When the ice melts, will the water level in the glass rise, fall, or remain unchanged? Explain
7	State the condition to be followed to attain thermal equilibrium.
8	Define an isolated system.Find out the change in internal energy of an isolated system
9	Distinguish between reversible and irreversible process
10	Define thermal efficiency of a heat engine?

Section B

All Questions can be answered. Each Question carries 6 marks (Ceiling : 36 Marks)

11	Explain stress-strain diagram for a ductile metal under tension .
12	Find out the gravitational torque on an object of arbitrary shape
13	Prove the equation of continuity for compressible and incompressible fluid.
14	Derive the Work done by an ideal gas in an adiabatic process
15	Prove that molar heat capacity of an ideal gas at constant pressure (*Cp*)is *greater* than the molar heat capacity at constant volume (C_V)
16	Explain the adiabatic processes for an ideal gas with P-V diagram
17	A Carnot engine whose high-temperature reservoir is at 620 K takes in 550 J of heat at this temperature in each cycle and gives up 335 J to the low-temperature reservoir. What is (a) the temperature of the low-temperature reservoir; (b) the thermal efficiency of the cycle
18	One kilogram of water at 0^0C is heated to 100^0C . Compute its change in entropy. Assume that the specific heat of water is constant at 4190 J/kg .K over this temperature range.

Section C		
Answer any ONE .Each Question carries 10 marks (1x10=10 Marks)		
19	Derive Bernoulli's equation, by applying the work-energy theorem to the fluid in a section of a flow tube.	
20	Define entropy and its relation with randomness.Show that the total entropy change during any reversible cycle is zero	